Week 5 — Confidence intervals

Business Forecasting

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Where we are

C1 Framing → C2 Describing data → C3 Uncertainty → C4–C5 Regression → C6 Forecasting

We are in **C3 — uncertainty**. A point estimate (a sample mean) is one number; this week we put a **range of plausible values** around it.

- **This week (W5): confidence intervals** — for a **mean** (normal & Student's t) and for a **variance** (chi-square).

Fast & hands-on: every interval we derive, you build in R with `qnorm` / `qt` / `qchisq` — practice in `W05_learnr.Rmd`.

Confidence Intervals

Confidence Intervals

- We calculated the mean price in our sample
- How confident are we that our estimate is close to the true parameter?
- A confidence interval measures the uncertainty around the estimate — a range of plausible values for the parameter.

Confidence Intervals

- Mean price in our sample was 1245.43
- Is it reasonable to think the true average price is 1100? What about 2000?
- Suppose we computed the confidence interval to be:

$$\{1086.64,\ 1404.22\}$$

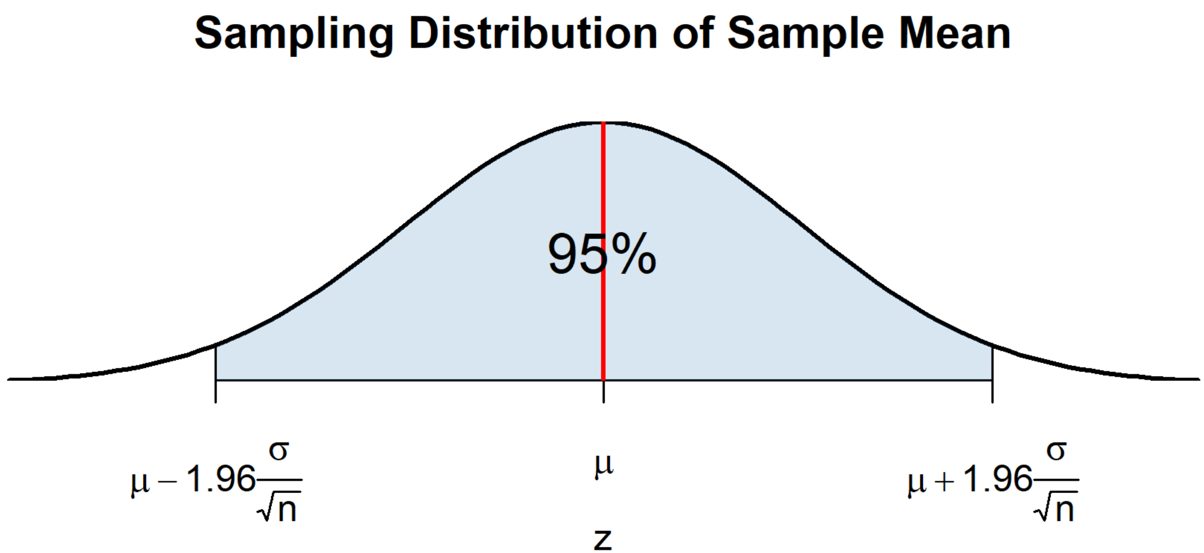
- Where do these numbers come from?

    1. The sampling distribution of the sample mean tells us how far a point estimate typically lands from the true mean.
    2. The confidence interval uses that to say where the true mean plausibly lies.

Sampling distribution

Q: How likely is it that a sample mean is far from the true mean?

- Recall the sampling distribution: $\bar{x} \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$
- Across repeated samples, 95% of the means land in the shaded region. Why 1.96?



Where 1.96 comes from

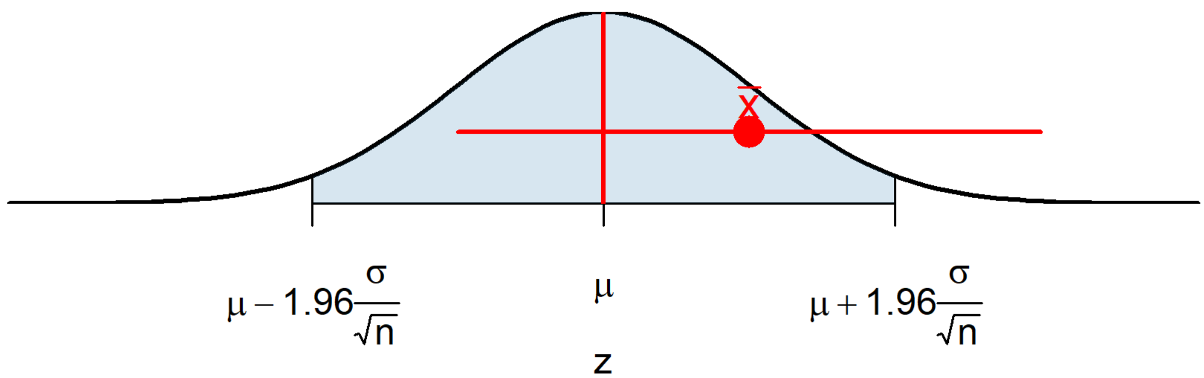
$$0.95 = P(-1.96 < Z < 1.96) = P\left(z_{\frac{\alpha}{2}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{1-\frac{\alpha}{2}}\right)$$
$$= P\left(\mu + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \bar{X} < \mu + z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right)$$

- The CLT guarantees $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is standard normal.
- What if you don't know σ? In large samples $s \to \sigma$, so $\frac{\bar{X}-\mu}{s/\sqrt{n}} \to N(0,1)$.
  - You may need a somewhat larger n so that s approximates σ well.

Margin of error

- So 95% of sample means fall within $1.96\frac{\sigma}{\sqrt{n}}$ of the true mean.
- This half-width is the margin of error: $\mathrm{ME} = z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$.
- The interval $\bar{x} \pm \mathrm{ME}$ covers μ whenever $|\mu - \bar{x}| < \mathrm{ME}$.

▶ learnr `W05_learnr.Rmd` → Exercise 4: Margin of error and how it shrinks with n — `qnorm(0.975) * se`, and 4× the sample halves it.



95% coverage across repeated samples

- Draw many samples, compute $\bar{x}$ and its interval for each.
- **95% of those intervals cover the true mean.** That is what "95% confidence" means.

Source: [Seeing Theory]

Calculation procedure

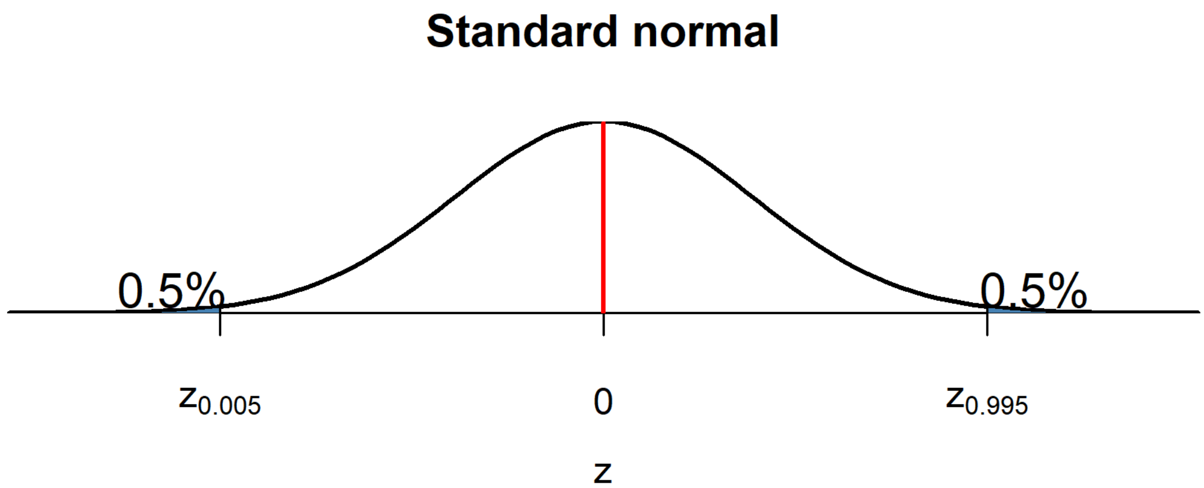
Use this when $n > 40$:

1. Take an IID sample.
2. Compute the mean $\bar{x}$ and standard deviation s.
   - Standard error = SD of the estimator: $SE = \frac{s}{\sqrt{n}}$.
3. Pick a confidence level $1 - \alpha$ (usually 90 / 95 / 99%).
   - α = probability of a Type 1 error. Example: 95% $\rightarrow \alpha = 0.05$.
4. Find critical values $z_{\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}$ with $P(z_{\frac{\alpha}{2}} < Z < z_{1-\frac{\alpha}{2}}) = 1 - \alpha$.
   - Example: 95% $\rightarrow z_{0.975} = 1.96$.
5. Build the interval: $\left\{\bar{x} - z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}, \quad \bar{x} + z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}\right\}$

▶ learnr `W05_learnr.Rmd` → Exercises 1–2: Point estimate, spread & standard error — `mean(x)`, `sd(x)`, and `se <- sd(x)/sqrt(n)`.

Finding critical values

- Suppose the confidence level is 99%, so $\alpha = 0.01$.
- We need $z_{1-\frac{\alpha}{2}}$ with $P(z_{\frac{\alpha}{2}} < Z < z_{1-\frac{\alpha}{2}}) = 0.99$.
- Equivalently $z_{0.995}$ is the 99.5% quantile of the standard normal → $z_{0.995} = 2.58$.



▶ learnr `W05_learnr.Rmd` → Exercise 5: Change the confidence level (90% and 99%) — find critical values with `qnorm()` and watch them grow.

Constructing a CI: worked example

90% CI for the average price of listings with cleanliness score > 4.5:

1. IID sample, $n = 100$ ✓
2. $\bar{x} = 1245.43$ and $s = 961.9$
3. Confidence level 90% $\rightarrow \alpha = 0.1$
4. Critical value: $z_{0.95} = 1.65$
5. Interval:

$$\left\{1245.43 - 1.65 \cdot \frac{961.9}{\sqrt{100}},\quad 1245.43 + 1.65 \cdot \frac{961.9}{\sqrt{100}}\right\} = \{1086.64,\ 1404.22\}$$

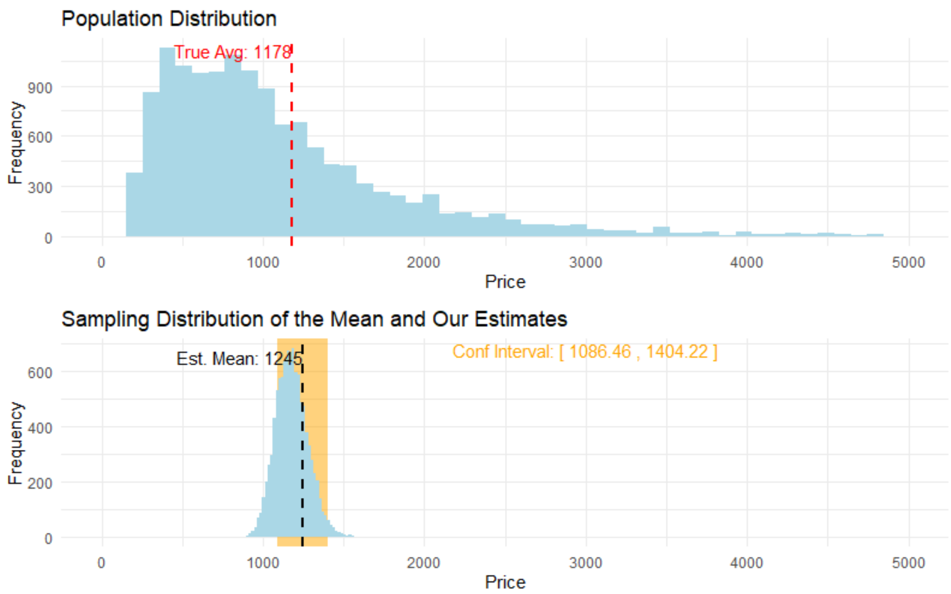
▶ learnr `W05_learnr.Rmd` → Exercise 3: A 95% confidence interval (normal) — `mean(x) + c(-1, 1) qnorm(0.975) se`.

Interpreting a confidence interval

$$CI_{90} = \{1086.64,\ 1404.22\}$$

- Correct: we are 90% confident the interval captures the true mean — i.e. that the true mean price is between 1086.64 and 1404.22.
- Incorrect: "with 90% probability the true mean is in this interval."
    - The computed interval is fixed and the true mean is fixed — no randomness left to attach a probability to.
    - The interval is random only before we draw the sample. After computing, the true mean is either in it or not.

▶ learnr `W05_learnr.Rmd` → Exercise 8: Interpreting a confidence interval — pick the correct "95% confident" statement.


Population Distribution
True Avg: 1178
Frequency
900
600
300
0
0
1000
2000
3000
4000
5000
Price
Sampling Distribution of the Mean and Our Estimates
Est. Mean: 1245
Conf Interval: [1086.46 , 1404.22]
Frequency
600
400
200
0
0
1000
2000
3000
4000
5000
Price

Shape of confidence intervals

$CI = \left\{\bar{x} \pm z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}\right\}$ is wider when:

- the confidence level is higher (99% is wider than 90%);
- the sample size n is smaller;
- the standard deviation σ is larger.

▶ learnr `W05_learnr.Rmd` → Exercises 4–5: Margin of error & confidence level — the width shrinks with larger `n` and grows with the confidence level.

When do we use *normal* critical values?

1. X is not normal:

- $n > 30$ — use the normal (use $n > 40$ if σ unknown, so s approximates σ).
- $n < 30$ — normal approximation is unreliable; you need specialized methods (e.g. bootstrap).

2. X is normal:

- Know $\sigma \rightarrow$ use the normal, any n (with σ, $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$).
- Don't know σ but $n > 40 \rightarrow$ normal (CLT kicks in).
- Don't know σ and $n < 40 \rightarrow$ use Student's t (s is a poor estimate of σ at low n).

What's Student's t?

If $X_1, \ldots, X_n$ are i.i.d. from $N(\mu, \sigma)$, then

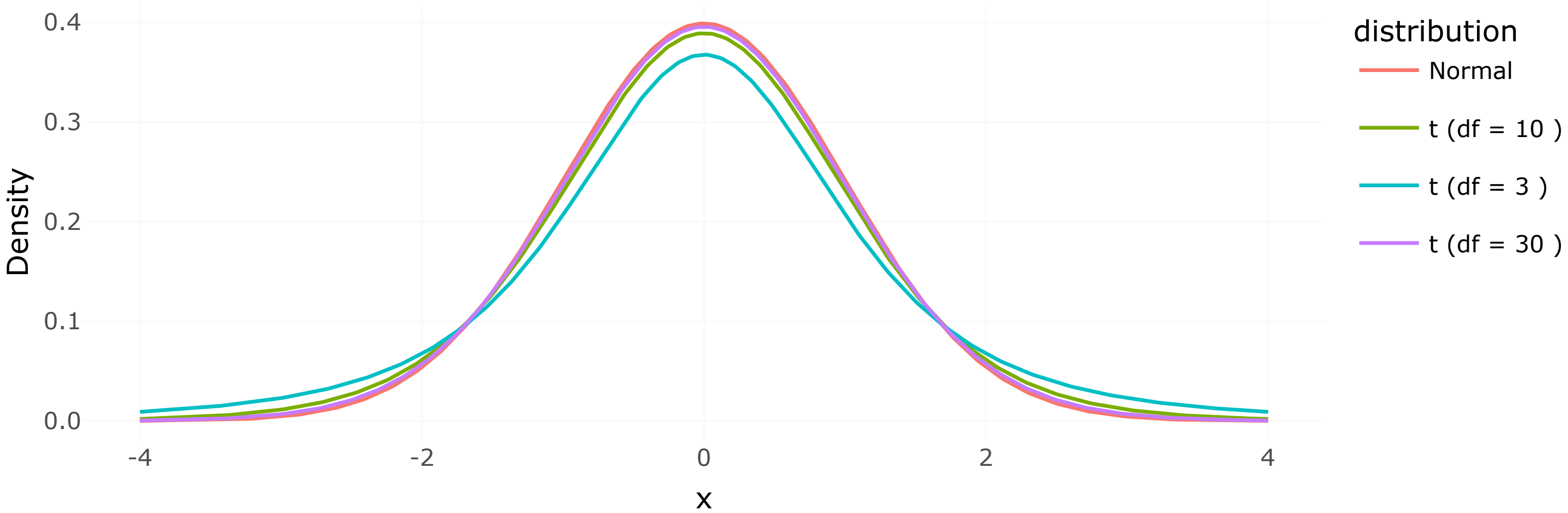
$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

with s the sample standard deviation, follows a Student's t distribution with $n - 1$ degrees of freedom:

$$T \sim t_{n-1}$$

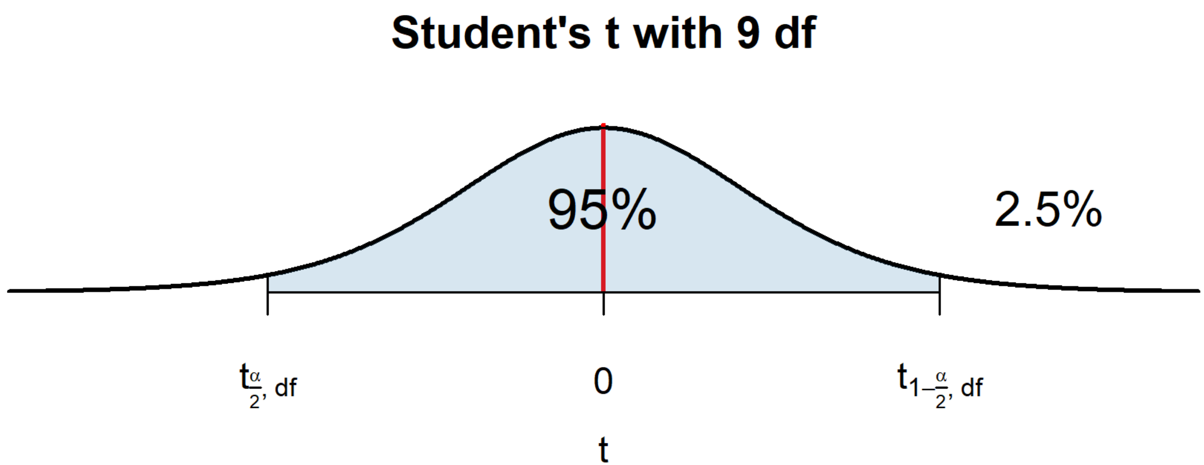
What's Student's t?

- Bell-shaped and symmetric around 0.
- Heavier tails than the normal — more uncertainty, because we don't know σ.
- Shape set by the degrees of freedom: as n grows, $t_{n-1} \to N(0,1)$ (CLT again).



Student's t critical values

1. Get the degrees of freedom: $n-1$.
2. Get the confidence level $1-\alpha$, hence $\alpha/2$.
3. Find $t_{1-\frac{\alpha}{2},n-1}$ with $P(T < t_{1-\frac{\alpha}{2},n-1}) = 1-\frac{\alpha}{2}$.



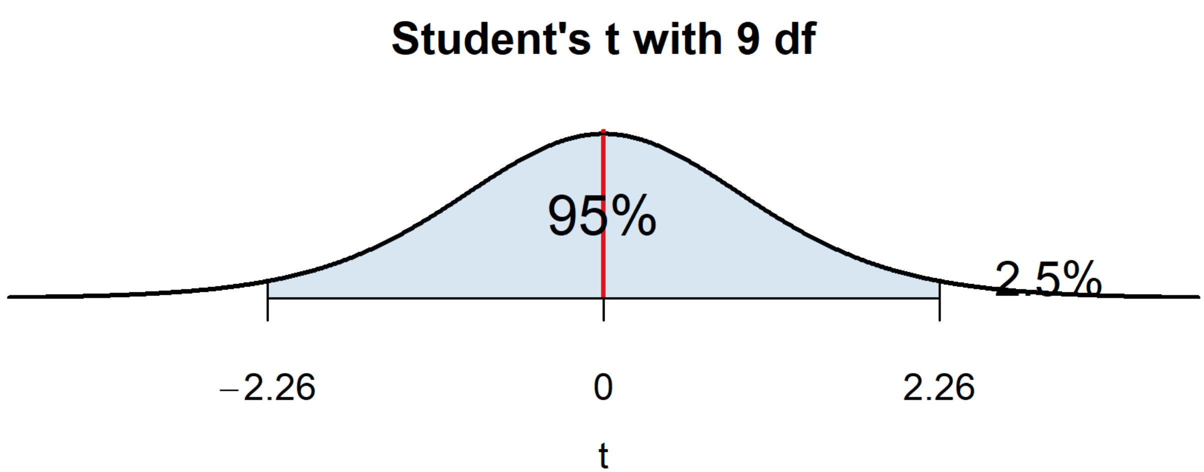
In R: `qt(0.975, 9)` for a 95% CI with 9 df.

t critical value: example

- $n = 10 \rightarrow df = 9$; confidence level 95% $\rightarrow \frac{\alpha}{2} = 0.025$.
- $t_{0.975,9} = 2.26$, then build the CI as before:

$$\left\{ \bar{x} - t_{1-\frac{\alpha}{2},n-1} \cdot \frac{s}{\sqrt{n}}, \quad \bar{x} + t_{1-\frac{\alpha}{2},n-1} \cdot \frac{s}{\sqrt{n}} \right\}$$

> ▶ learnr `W05_learnr.Rmd` → Exercise 7: Read the CI straight off `t.test()` — `t.test(x)` prints (and returns) the 95% CI.



Practice: free-shipping CI

Free shipping was offered to a random group of customers. Did it raise spending? Spend per customer ($n = 10$):

$$157.80,\ 192.45,\ 210.20,\ 175.60,\ 198.30,\ 180.90,\ 205.75,\ 185.20,\ 177.40,\ 195.60$$

a) Compute the 90% CI. What assumptions do you need (small $n \rightarrow$ Student's t)?

b) Average spending **without** free shipping is \$182. Can you conclude free shipping changed spending?

> ▶ learnr `W05_learnr.Rmd` → Exercise 6: Small-sample t interval — build this small-sample CI with `qt(0.975, df = n - 1)`.

EXAM QUESTION — Fall 2025 Midterm 1

Real questions from last year's Midterm 1 — Households & Environment Survey ($n = 14{,}505$ households, each surveyed once, 2017). We solve these together in class using `households_environment.rda` (object `exam_data`), in your Week 5 folder.

> **Q7.** A transportation agency wants to estimate average weekly household gasoline spending ($n = 14{,}505$). Construct a 90% confidence interval for the mean and report the lower bound.

> **Q8.** What happens to the interval if you change the confidence level to 99%? (no calculation)

Large $n \to$ use the normal critical value. Higher confidence $\to$ wider interval, so a lower lower bound.

> **Practice now (this week's pack):** the CI-construction and confidence-level problems in the Week 5 set, then re-run Q7 on `exam_data`.

What you must be able to do this week

- Construct a confidence interval for a mean — compute $\bar{x}$, s, the standard error, and the margin of error, then $\bar{x} \pm \mathrm{ME}$.
- Choose the right critical value: normal (`qnorm`, large n) vs Student's t (`qt`, small n, σ unknown).
- Interpret a CI correctly — confidence about the procedure, *not* a probability about a fixed interval.
- Explain what makes an interval wider or narrower: confidence level, sample size, and spread.
- Build a confidence interval for a variance with the (asymmetric) chi-square quantiles (`qchisq`).

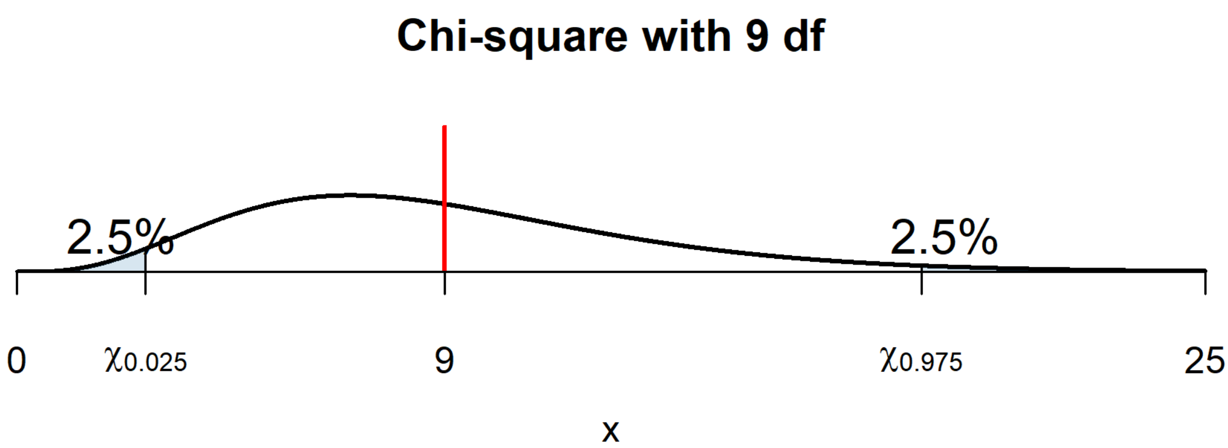
Next: from estimating parameters to testing claims about them — hypothesis testing.

Appendix — optional theory refresher

Confidence intervals for variance

- Sometimes we care about spread, not the mean.
- A machine making bolts with high diameter variance is unreliable even if the average is correct.
- For $X_1, \ldots, X_n$ from a normal, the sample variance $S^2 = \frac{\sum_i (X_i - \bar{X})^2}{n-1}$ satisfies:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad P\left(\chi^2_{0.025,n-1} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{0.975,n-1}\right) = 0.95$$



CI for variance: construction

Invert the chi-square statement to isolate σ^2:

$$\begin{aligned} 0.95 &= P\left(\chi^2_{0.025,n-1} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{0.975,n-1}\right) \\ &= P\left(\frac{(n-1)S^2}{\chi^2_{0.975,n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{0.025,n-1}}\right) \end{aligned}$$

So, generally:

$$CI_{1-\alpha} = \left\{ \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}, \ \frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}} \right\}$$

- The χ^2 quantiles come from R via `qchisq(alpha, df)`. Note the interval is asymmetric (chi-square is skewed).

Practice: sausage-variance CI

Quality control: you measure fat content in sausages. From a random sample of **12** sausages you find a sample variance of **20 grams²**. Find the **99% confidence interval for the population variance**. What assumptions do you need?

> ▶ learnr Bonus practice — use `qchisq()` for the two critical values and build this variance interval. See `W05_learnr.Rmd`.